



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECP3744 - 01 ENTERPRISE SYSTEM DESIGN AND MODELLING (WBL)

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SAP BTP REPORT

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Introduction

This report discusses the creation of a business-friendly chatbot based on SAP Conversational AI, specifically designed to work with the logistics and shipping business. This solution is also capable of processing complex, transactional user intents in contrast to simple automated systems, including real-time package statuses and shipping price quotes. Combining the innovative natural language processing and the functional back-end features, the project shows how AI can be turned into a simple interface and a powerful working tool.

The project will be divided into three strategic stages, the first one being the development of a logic-heavy chatbot that uses webhooks to retrieve dynamic data to ship customers. After the creation of these main features, is the deployment of the web, where it is embedded in a professional site, but the visual identity and user experience are preserved. Lastly, the project discusses the application of an FAQ chatbot, which shows how to use structured data in CSV files to automate the response to frequently asked questions. Overall, these modules offer a detailed roadmap to the creation, deployment, and management of enterprise-grade conversational agents that will simplify the communication process and increase customer satisfaction.

Tutorials

1. Create Chatbot for Customers of Shipping Company

1.1 Create skeleton bot

Click the New bot and enter all the bot details such as name, description, language, data policy, and bot visibility.

3

Create your bot

thecodester / ShippingBot

Your new bot will be created as: **shippingbot**

Description (optional)

A bot to help shipping customers track and price packages

Topics (optionally categorize your bot with up to 6 topics to improve your bot training)

Add topics

Try: [Customer support](#), [Food and beverage](#), [Telecommunications](#)

Default language (your bot is multilingual, you can add more languages later on)

 English Advanced

1.2 Create entity for parcel number

Data extracted from a conversation is defined as entities. The chatbot must be able to extract the parcel number from within the conversation as following the tracking number based on one of UPS's standards.

Create a custom entity

thecoderster / shippingbot-new / PARCEL-NUMBER

Your new entity will be created as: **parcel-number**

Ⓢ HOW DO CUSTOM ENTITIES WORK?

☐ **Restricted entity**

Useful if you have a strict list of words to detect and don't need automatic detection of entity.

#PIZZA

- ☐ cheese
- ☐ calzone
- ☐ pepperoni

OR

☐ **Free entity**

Useful if you don't have a strict list of values and want machine learning to detect all possible values.

#PHONE TYPE

- ☐ I want to buy an iPhone
- ☐ My Huawei is broken
- ☐ I'd like to have the new Samsung phone

OR

☒ **Regex entity**

Useful if you have a pattern to detect and don't need machine learning detection of the entity.

#BANK

/ [A-Z]{2,} /

My IBAN is 34294Z49390

Match 1
Full Match 3 - 7 IBAN

A detected entity with a confidence score strictly greater than the matching strictness is kept and returned in the JSON. Our recommendation is a matching strictness between 50 and 70. (Not applicable to regex entities)

Matching strictness 0 100

Cancel
CREATE

1.3 Add intent yes/no and tracking parcels.

Intents are the things people want to tell our bot. Within an intent, it indicates all the possible expressions that a person might use to communicate that intent. As SAP Conversational AI is collaborative, the intent yes/no and tracking parcel can get it by forking from an intent someone has already created and clone it right into our bot.

The image shows two side-by-side screenshots of the SAP Conversational AI 'Fork' dialog. Both dialogs are titled 'Where should we fork the intent @yes?' and 'Where should we fork the intent @track-parcel?'. Each dialog has a form with the following fields: 'In which bot?' (a dropdown menu with 'thecodester/shippingbot' selected), 'In which language?' (a dropdown menu with 'ALL' selected), 'Select your intent' (a dropdown menu with 'Select...' selected), and 'For identical expressions' (two radio buttons: 'Overwrite identical expressions in the existing intent' and 'Preserve identical expressions in the existing intent', with the latter selected). At the bottom right of each dialog are 'Cancel' and 'FORK' buttons.

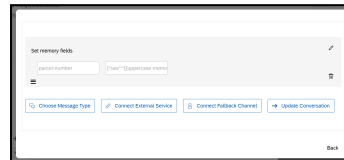
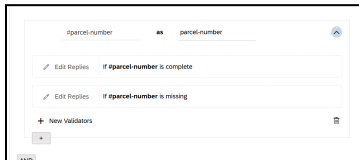
1.4 Create skill for tracking parcel

The skill is built so that the bot knows what to do in response to this intent. First, added a skill called track-parcel, then added a trigger so that this skill is triggered when the @track-parcel intent is detected.

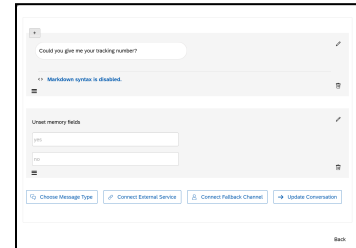
The image shows two side-by-side screenshots of the SAP Conversational AI interface. The left screenshot shows a grid of skills: 'track-parcel', 'price-parcel', 'greetings', 'fallback', and 'disambiguation'. The right screenshot shows the configuration for the 'track-parcel' skill. It has tabs for 'README.md', 'Triggers', 'Requirements', and 'Actions'. The 'Triggers' tab is selected, showing a trigger rule: 'If @track-parcel is-present'. There is a 'How do triggers work?' link and a '+' button to add more triggers.

1.5 Add requirement to skill (parcel-number)

The requirements for the track-parcel skill indicate the data the bot needs to extract from the conversation before executing the skill's action. In the requirements tab, the required information is set, which is the #parcel number as parcel-number. This is the name in the memory for storing the data for this entity. Then we expand the parcel-number, and set the actions taken if the parcel-number is complete or missing.



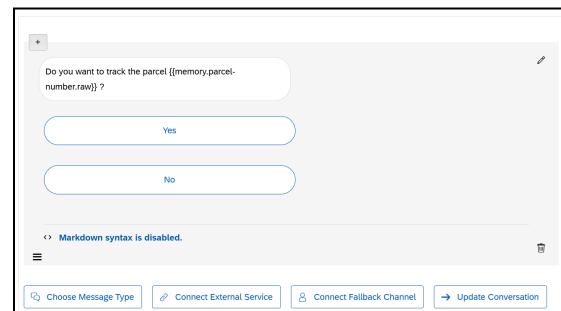
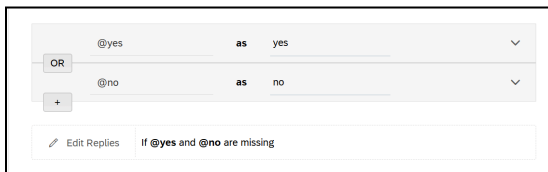
Parcel-number is complete



Parcel-number is missing

1.6 Add requirement to skill (yes/no)

A second set of requirements is added where it is the user's confirmation whether they want to track the package. Message replies when yes / no are missing is added and the code `{{memory.parcel-number.raw}}` means that it will retrieve the parcel number from the memory.



Replies when the yes / no are missing

1.7 Add action to skill

Action is executed when the skill is triggered

The action when @no skill is triggered

The action when @yes skill is triggered

1.8 Add Webhook to Chatbot to Retrieve Tracking Info

Default endpoint for webhooks is set in the setting under the bot builder field to the application on SAP Business Technology Platform.

1.9 Create entity for parcel size

Entity parcel size is created and the valid values are uploaded.

Entity Value	Selection	Edit	Delete
0-1kg	☐		
>10kg	☐		
1-3kg	☐		
3-10kg	☐		
extra large	☐		
l	☐		
large	☐		
m	☐		
medium	☐		
s	☐		
small	☐		
xl	☐		

1.10 Add intent for requesting shipping price

Intent is defined so that the bot can know when the customer is asking for pricing information. The intent is forked from the other peoples intent called `@rate-parcel` intent. Example of the intent:

☐	i want to ship the package to johor, malaysia. how much?			
☐	i want to send package to skudai. how much it cost?			
☐	how much to send package to selangor,malaysia?			
☐	how much to send package to johor,malaysia?			
☐	send to UTM Skudai			
☐	Eiffel Tower			
☐	rate-parcel			

1.11 Build skill for requesting shipping quote

The skills builded are called price-parcel skills. It makes the bot have a location and parcel size, makes an API call to get a map of the location and displays the map plus the price from memory based on the enrichments created earlier.

TrainBuildConnectMonitor

shippingbot / price-parcel  Business  

README.mdTriggersRequirementsActions

Trigger this skill only if:  How do triggers work?



If @rate-parcel is-present  



1.12 Add requirements to skill

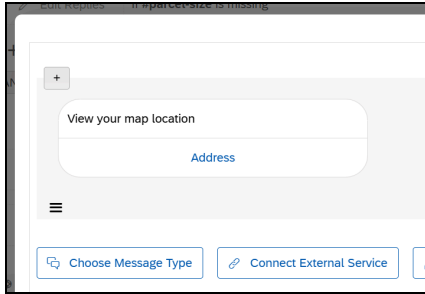
In the requirements subtab, requirements as follows were added and the replies for each requirement is added.

The screenshot shows a requirements subtab interface with a British flag icon in the top left. The interface is organized into a logical structure using AND and OR operators. The first section is connected by an AND operator and contains two requirements: '#parcel-size as parcel-size' and '#malaysia-places as malaysia-places'. Below this is a plus sign button. The second section is also connected by an AND operator and contains two requirements connected by an OR operator: '@yes as yes' and '@no as no'. Below this is another plus sign button. At the bottom, there is an 'Edit Replies' button and a text box containing the instruction 'If @yes and @no are missing'. A final plus sign button is located at the very bottom of the interface.

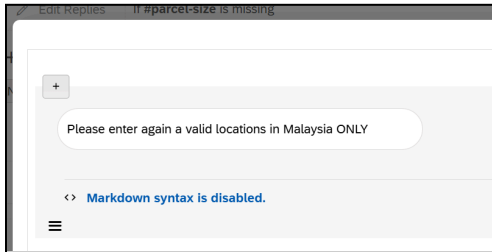
If the intent does not mention the size, the requirement parcel-size is missing and it will display the sizes that user can choose.

The screenshot displays a selection screen with four distinct size options arranged in a 2x2 grid. Each option includes a title, a weight range, and a selection button. The options are: 'Small' (0-1kg) with a 'Choose S' button, 'Medium' (1-3kg) with a 'Choose M' button, 'Large' (3-10kg) with a 'Choose L' button, and 'Extra Large' (>10kg) with a 'Choose XL' button. Each option is presented in a light gray box with a white header area for the text.

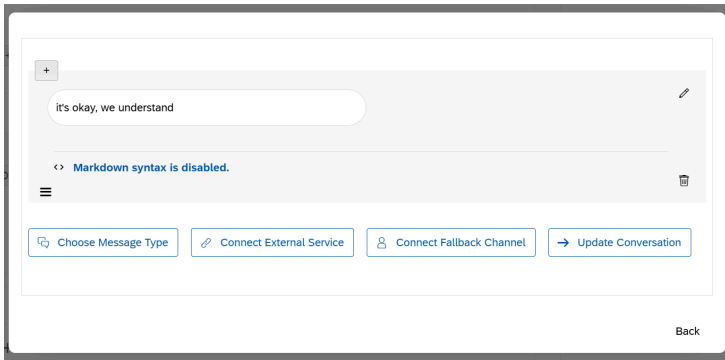
If the location is mentioned completely, the bot will display the maps from the location retrieved.



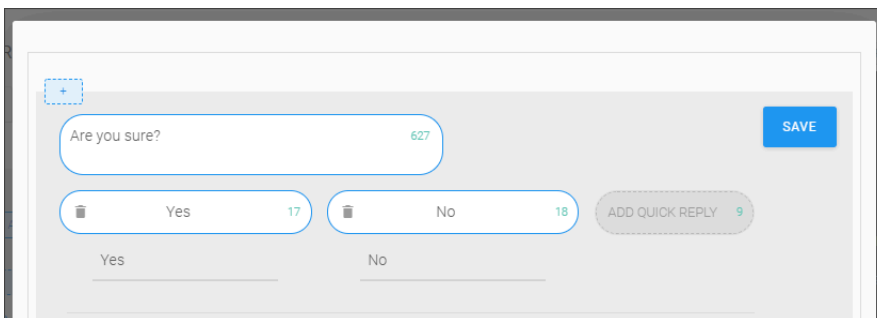
If the location is missing, the bot will ask the user to enter back the location.



If the user are not sure with the question, and clicks no, message replies will appear.



Then, the bot will display confirmation when the yes/no is missing



1.13 Display results

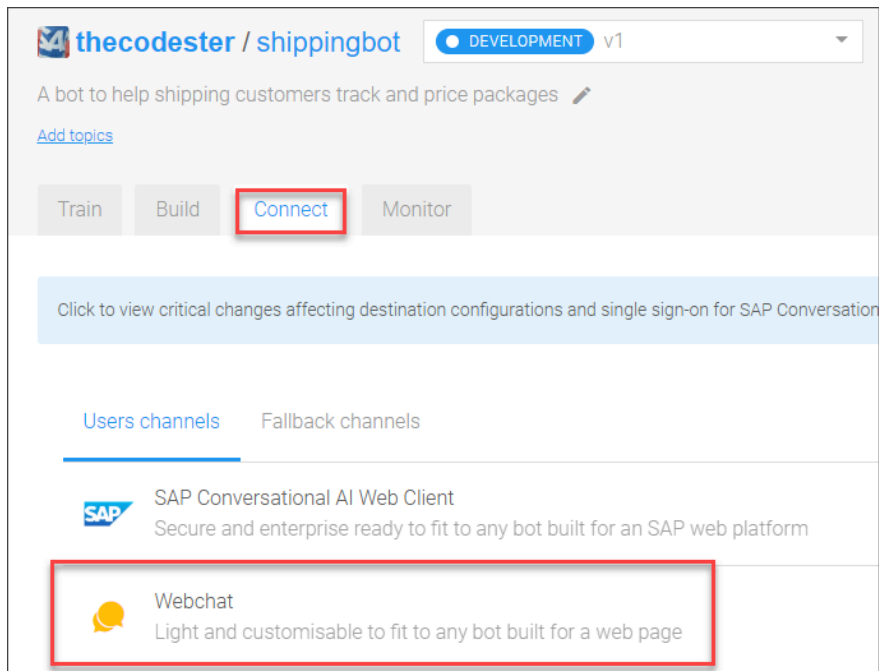
In the action tab, when the customer said yes, the results will be displayed.

The screenshot shows a configuration interface for a chatbot. At the top, there is a header with a flag icon. Below it, a conditional trigger is set: "If `_memory.yes` is-present". To the right of this trigger are icons for deleting and adding more conditions. Below the trigger, there is a large text area containing a message template: "A `{{memory.parcel-size.name}}` parcel sent to `{{memory.location.formatted}}` will cost `${{memory.parcel-size.price}}`". To the right of this text area is an edit icon. Below the text area, there is a small blue link that says "Markdown syntax is disabled." with a trash icon to its right. At the bottom of the configuration area, there is a button labeled "Reset all memory" with an edit icon to its right. The entire configuration area is enclosed in a light gray box with a plus icon in the top left corner.

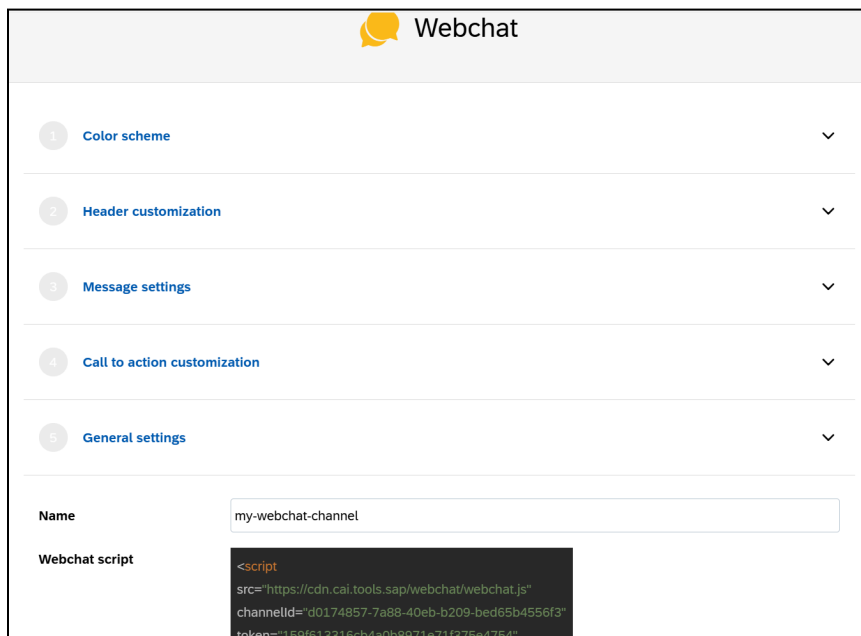
If no, it will reset the memory

The screenshot shows a configuration interface for a chatbot. At the top, there is a header with a flag icon. Below it, a conditional trigger is set: "If `_memory.no` is-present". To the right of this trigger are icons for deleting and adding more conditions. Below the trigger, there is a large text area containing a message template: "Reset all memory". To the right of this text area is an edit icon. At the bottom of the configuration area, there are four buttons: "Choose Message Type", "Connect External Service", "Connect Fallback Channel", and "Update Conversation". The entire configuration area is enclosed in a light gray box with a plus icon in the top left corner.

2. Deploy an SAP Conversational AI Chatbot on a Web Site



Chatbot interface is set by following the preference.



Webchat script is copied into the iisstart.html under the Microsoft IIS web server.

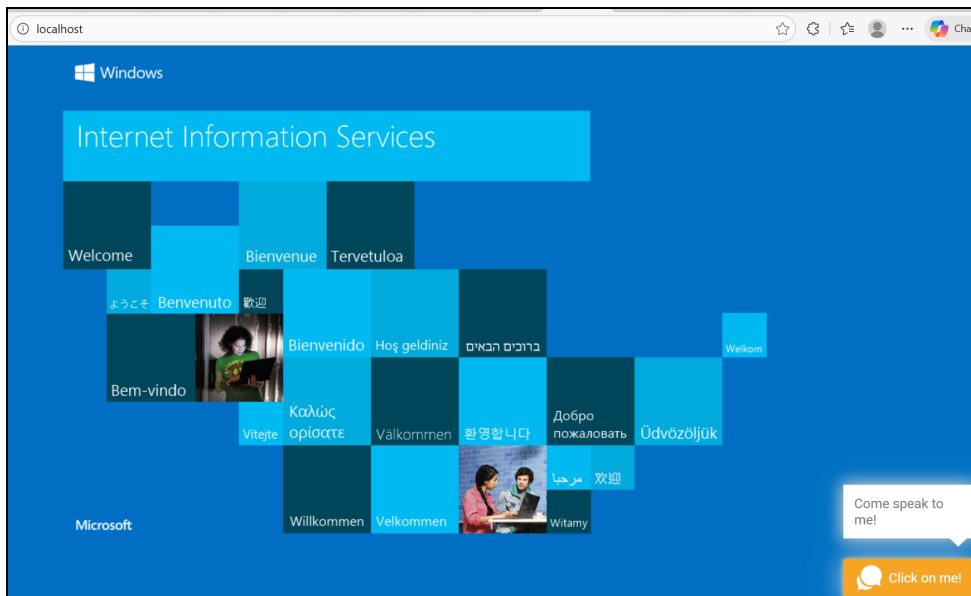
```
Untitled iisstart.htm x +
File Edit View
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1" />
<title>IIS Windows</title>
<style type="text/css">
<!--
body {
    color:#000000;
    background-color:#0072C6;
    margin:0;
}

#container {
    margin-left:auto;
    margin-right:auto;
    text-align:center;
}

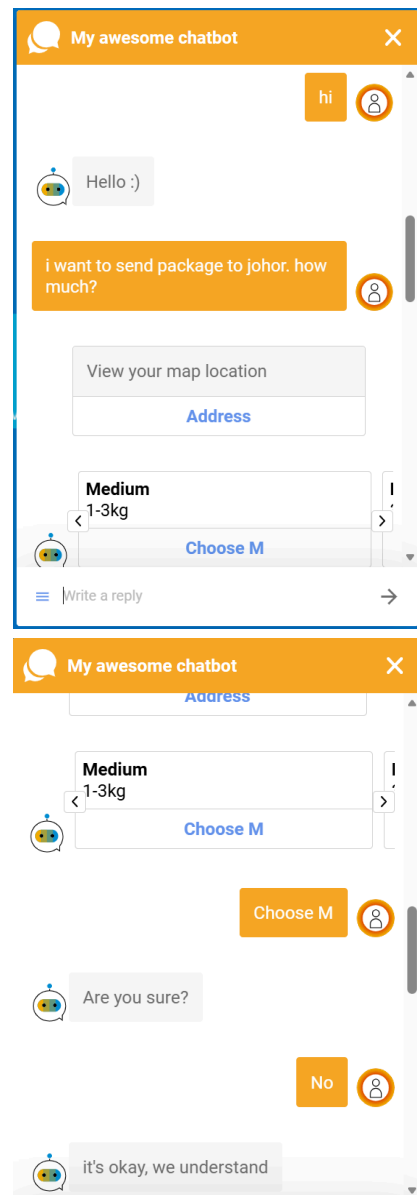
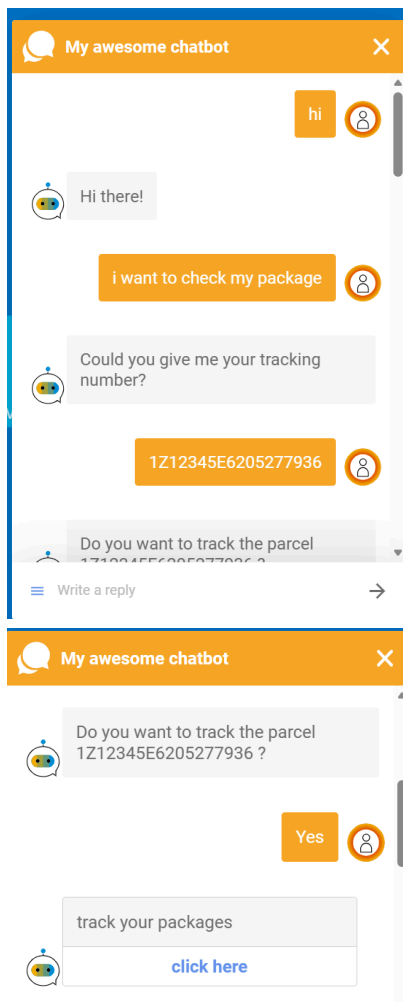
a img {
    border:none;
}

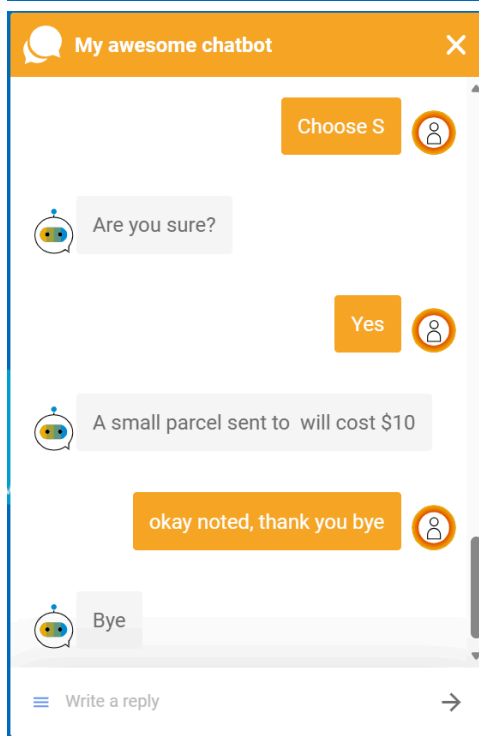
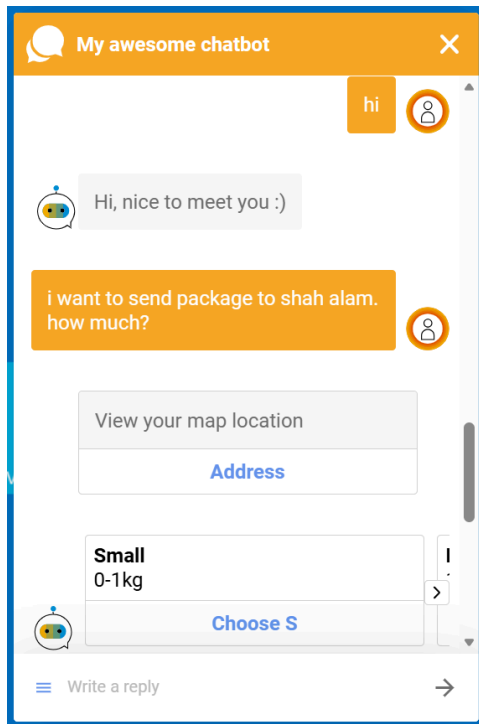
-->
</style>
</head>
<body>
<script
src="https://cdn.cai.tools.sap/webchat/webchat.js"
channelId="d0174857-7a88-40eb-b209-bed65b4556f3"
token="159f613316cb4a0b8971e71f375e4754"
id="cai-webchat">
</script>
<div id="container">
<a href="http://go.microsoft.com/fwlink/?linkid=66138&clid=0x409"></a>
</div>
</body>
</html>
```

Now, chatbot is applied to Microsoft IIS web server.



2.1 Conversation Example:





3. Build First FAQ Chatbot

3.1 Create a skeleton FAQ bot and all the details are inserted. Then csv file is uploaded where it includes questions and answers.

1 What do you want your chatbot to do?


Perform Actions
using conversational flows


Retrieve Answers
from FAQ documents

2 Create your bot

thecodester / FAQBot

Your new bot will be created as: **faqbot**

Create and manage your FAQ documents

 + Add Language

Your documents 1  FAQ Bot Creation Guide

+ New Document

Document Name

 **howtobuildachatbot**

Source: CSV 113  

3.2 The skills

Train Build Connect Monitor

© Settings

Build your bot's skills 

Your skills

Skill Name	Active
faq	☒
customer-satisfaction-prompt	☐
customer-satisfaction-reply	☐
small-talk	☐

3.3 Monitor and improve the bot

In actions tab, it can monitor what questions people are asking and either can add answers for these questions or add these questions to existing answers

The screenshot shows the 'Monitor' tab in the bot interface. At the top, there are tabs for 'Train', 'Build', 'Connect', and 'Monitor', with 'Monitor' being the active tab. Below these are 'Log Feed', 'Usage Metrics', and 'Conversation Logs'. The main heading is 'Discover all sentences analyzed by your bot', with a '50 per page' dropdown. Below this are filters: 'Using Log Feed', 'Assign Feature', 'Using Confidence Score', and 'Measuring Accuracy'. A 'Filters' button is also present. The 'User logs' section shows a table of user queries. The first three rows are highlighted in blue. The first row shows a query 'How do I use intents?' with a 'MATCHED' status of 100. The second row shows the same query with a 'MATCHED' status of 74. The third row shows a query 'What is a skill?' with a 'MATCHED' status of 100. A 'Download Logs' button is at the bottom right.

Log Feed	Usage Metrics	Conversation Logs
Discover all sentences analyzed by your bot		
50 per page		
Using Log Feed Assign Feature Using Confidence Score Measuring Accuracy		
Filters		
User logs		
All Refresh		
☐	howtobuildchatbot	Today 12/30/25 at 07:10:22 How do I use intents? MATCHED 100
☐	howtobuildchatbot	Today 12/30/25 at 07:07:08 How do I use intents? MATCHED 74
☐	howtobuildchatbot	Today 12/30/25 at 07:04:19 What is a skill? MATCHED 100
Download Logs		

This screenshot shows the 'Monitor' tab with a red box highlighting the 'Monitor' tab and the 'Log Feed' section. The 'Log Feed' section shows a table of user queries. The first row is highlighted in blue. The first row shows a query 'How do I use intents?' with a 'MATCHED' status of 100. The second row shows the same query with a 'MATCHED' status of 74. The third row shows a query 'What is a skill?' with a 'MATCHED' status of 100. A 'Download CSV' button is at the bottom right.

Log Feed	Usage Metrics	Conversation Logs
Discover all sentences analyzed by your bot		
50 per page		
Using Log Feed Assign Feature Using Confidence Score Measuring Accuracy		
Filters		
User logs		
All Refresh		
☐	howtobuildchatbot	Today 12/30/25 at 07:10:22 How do I use intents? MATCHED 100
☐	howtobuildchatbot	Today 12/30/25 at 07:07:08 How do I use intents? MATCHED 74
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☐	howtobuildchatbot	Today 12/30/25 at 07:04:19 What is a skill? MATCHED 100
Download CSV		

Conclusion

By completing these modules, I have gained a comprehensive understanding of the end-to-end lifecycle of a conversational AI agent, from initial intent mapping to live web deployment. I can now confidently implement these into subsystems in future projects such as in my staff scheduling subsystem.

Furthermore, I recognize that the deployment of a chatbot is only the beginning. To ensure long-term accuracy and user satisfaction, a "smart" chatbot requires a rigorous continuous improvement cycle. This involves frequently analyzing conversation logs to identify "fallback" scenarios where the bot failed to understand a query

Reference

<https://developers.sap.com/mission.conversational-ai-business-bots-1star.html>